Mines Paris PSL - Etilab

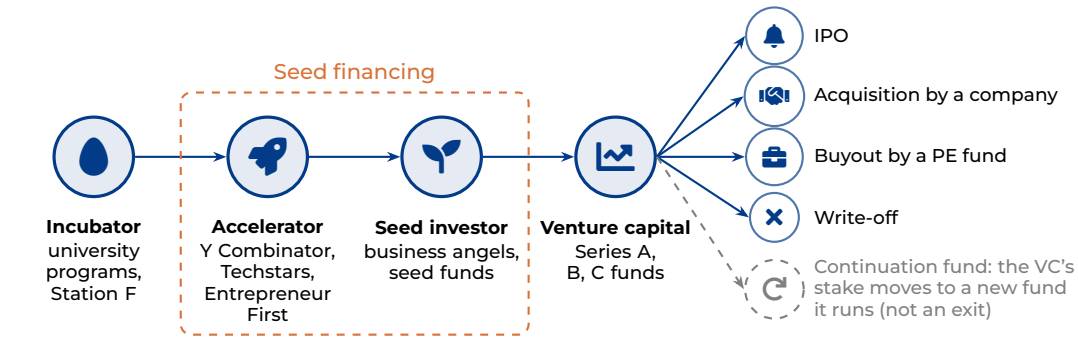
Protection and Dilution: SAFEs and MFN Clauses in Seed Financing

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Under the supervision of
Pierre Fleckinger and Philippe Mustar

November 3, 2026

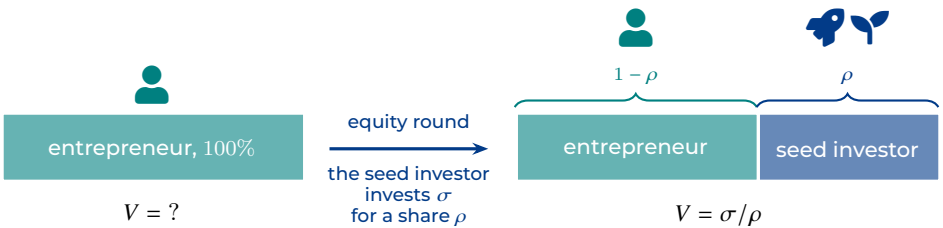
The life of a venture-backed startup

A startup raises money from a sequence of investors before it exits.



Exit routes follow Cumming and MacIntosh (2003): IPO, acquisition, secondary sale, buyback, write-off.

A first equity round combines the first issuance of N shares at price p.



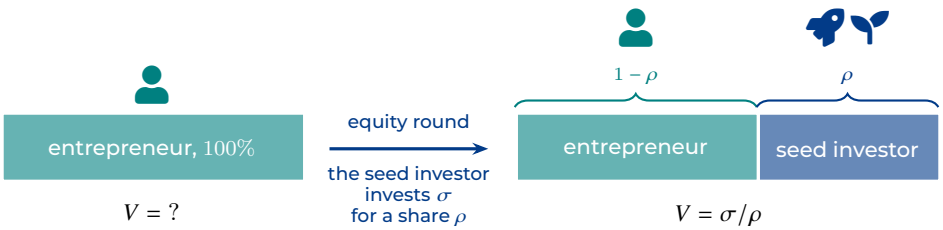
Definition

An equity round is characterized by (σ, ρ, N, p) such that $Np = \sigma/\rho = V$.

A seed investor invests \$120K for 7% equity, so $V = \$120\text{K}/0.07 \approx \$1.71\text{M}$.

Contracts in seed financing
The equity round

A first equity round combines the first issuance of N shares at price p.



Definition

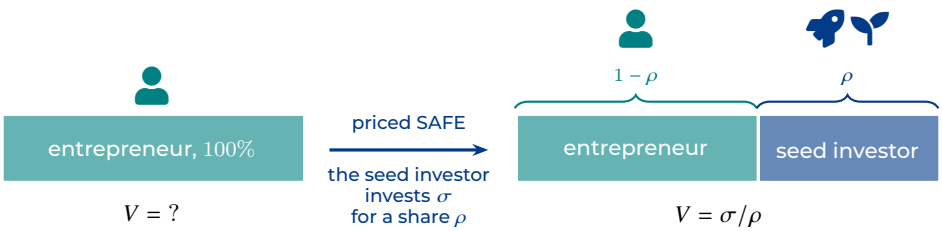
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But equity rounds are administratively heavy (new share class, legal fees).

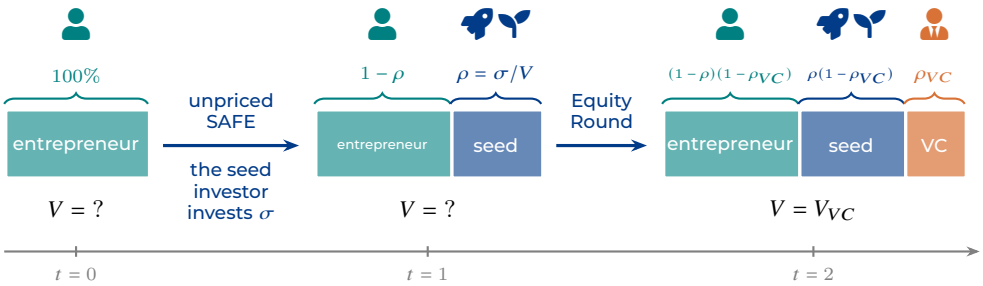
Contracts in seed financing
Priced SAFE (Simple Agreement for Future Equity)

A first priced SAFE defines the project's first valuation without issuance of shares.

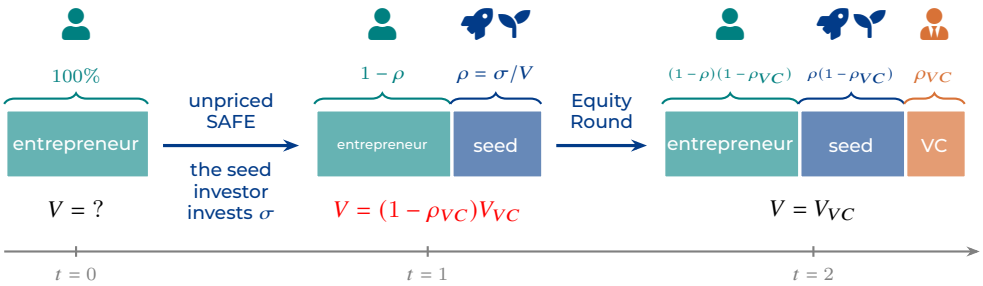


A seed investor invests \$125K for 7% on a SAFE, so $V = \$125\text{K}/0.07 \approx \$1.79\text{M}$.

A first unpriced SAFE with no cap and no discount leaves the valuation to the next round.

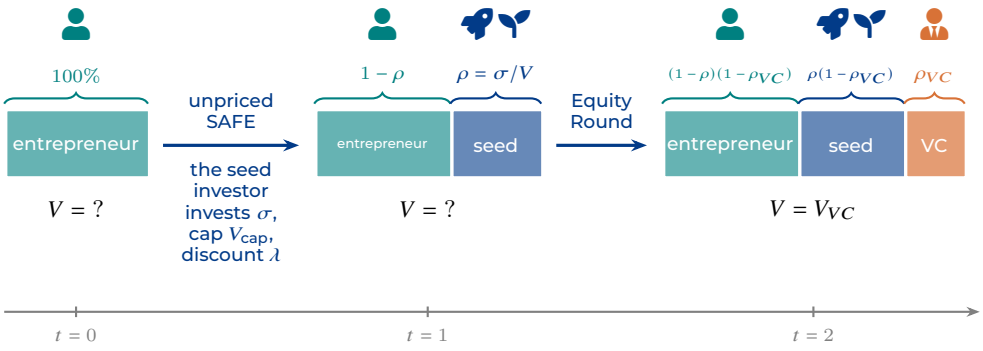


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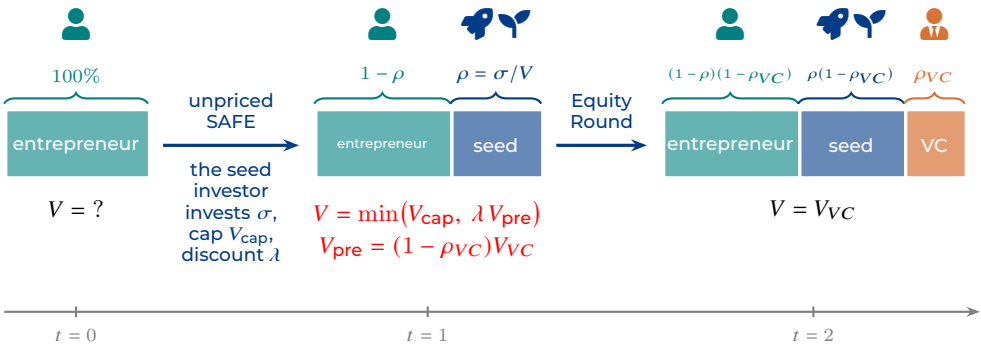
Contracts in seed financing
Unpriced SAFE with a cap V_{cap} and a discount λ

A first unpriced SAFE defines a valuation cap and a discount without issuance of shares.



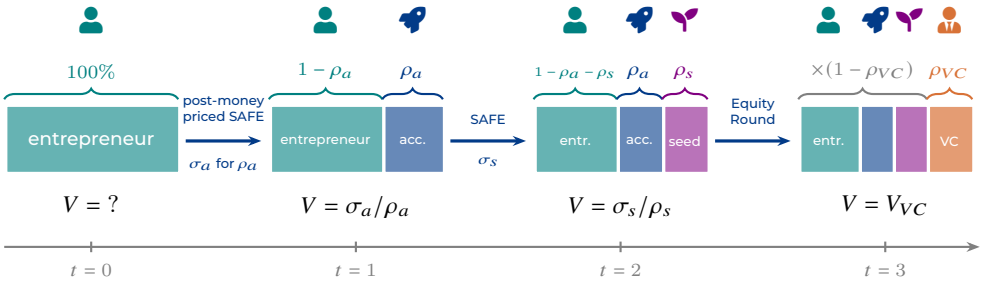
Contracts in seed financing
Unpriced SAFE with a cap V_{cap} and a discount λ

A first unpriced SAFE defines a valuation cap and a discount without issuance of shares.



Contracts in seed financing
The accelerator's post-money priced SAFE

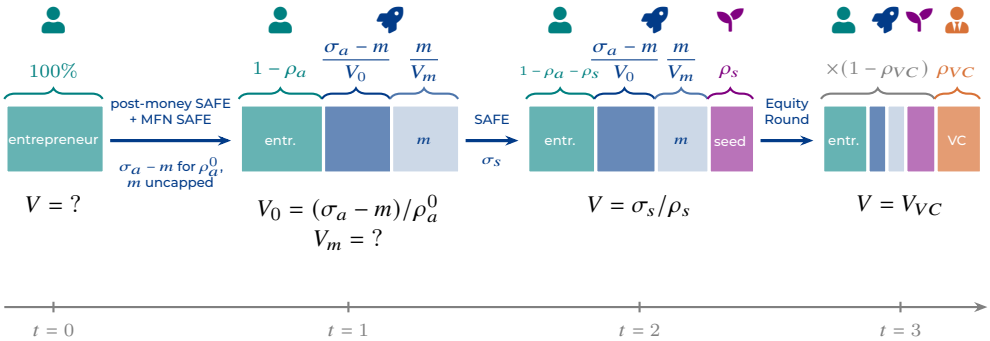
The accelerator's post-money priced SAFE fixes its share ρ_a until the equity round.



"Post-money" ensures that other SAFEs cannot dilute the accelerator's SAFE.

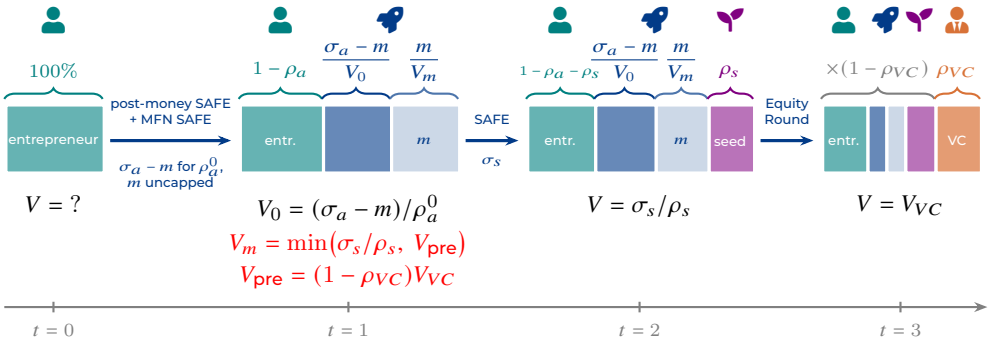
Contracts in seed financing
Post-money SAFE and MFN SAFE

The accelerator can add to its post-money SAFE an MFN SAFE, which converts on the best upcoming terms.



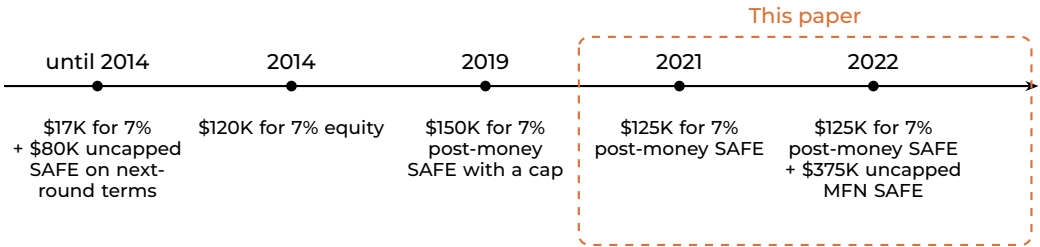
The accelerator's SAFEs convert at two valuations. The MFN SAFE buys m/V_m: it takes more from a seed round with a low valuation.

The accelerator can add to its post-money SAFE an MFN SAFE, which converts on the best upcoming terms.



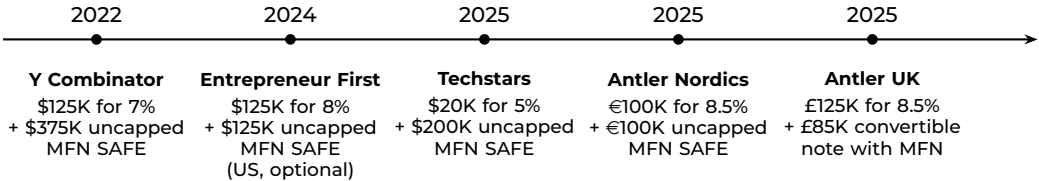
The accelerator's SAFEs convert at two valuations. The MFN SAFE buys m/V_m: it takes more from a seed round with a low valuation.

Chronology of Y Combinator deal terms



Accelerators' MFN SAFEs

Y Combinator launched a trend among accelerators.



A convertible note works like a SAFE, but it is debt: it carries an interest rate and a maturity, and ranks senior in liquidation.

The question

Research question

Is it worth adding an MFN SAFE to a post-money priced SAFE?

- For the accelerator: does the clause raise its profit, and how should it split its money between the two SAFEs?
- For the projects: does it raise the probability of a seed round? Of an equity round?
- Across projects: how do these anti-dilution clauses separate software and deep-tech projects?

Literature Review

Entrepreneurship as experimentation

- Kerr, Nanda and Rhodes-Kropf (2014): value is learned by investing.
- Nanda and Rhodes-Kropf (2017): financing risk shapes early rounds.
- Ewens, Nanda and Rhodes-Kropf (2018): cheaper experiments, more bets.
- Bergemann and Hege (2005): staged financing with learning.

Securities in venture capital

- Admati and Pfleiderer (1994): a fixed fraction of later rounds.
- Kaplan and Strömberg (2003): cash-flow and control rights.
- Coyle and Green (2014): capped notes, then SAFEs.

Investors before the VC

- Hellmann and Thiele (2015): angels depend on VCs.
- Hallen, Cohen and Bingham (2020): accelerators speed up ventures.
- Kim and Wagman (2014): an accelerator's portfolio and disclosure.
- Zarei, Rasti-Barzoki and Moon (2022): selection and mentorship.

Most-favored-customer clauses

- Cooper (1986): the clause keeps prices high.
- Besanko and Lyon (1993): a contemporaneous clause is weaker.
- McAfee and Schwartz (1994): no protection for early signatories.

To our knowledge, the first model of an accelerator's protection against seed dilution.

Findings

When the accelerator invests, adding an MFN SAFE to a post-money priced SAFE is worth it.

For the accelerator

- The MFN SAFE beats any fixed share: it takes more from small seed rounds.
- The seed investor answers by investing beyond its need on the smallest rounds.
- The clause covers at most the accelerator's contribution.

For the projects

- A protection shortens the seed investor's zone.
- The accelerator may invest more or less, which slides the zone.
- On small rounds, the seed investor invests more than needed, which helps projects reach the equity round.

Software and deep tech

- A seed round covers at most $W(I) < I$.
- Software: the accelerator invests, and all its money goes in the MFN SAFE.
- Larger needs: it invests nothing, and its fixed share costs about a quarter of the total surplus.

The agents

The project

- A liquidity shock $\ell \sim g$.
- A scaling cost I.
- A value $\pi \sim f$.

(A1) $E(\pi) < I$.

(A2) f, g positive, log-concave.

The accelerator

- Owns the whole project.
- Spends σ_a before ℓ and π.
- Can add anti-dilution clauses.

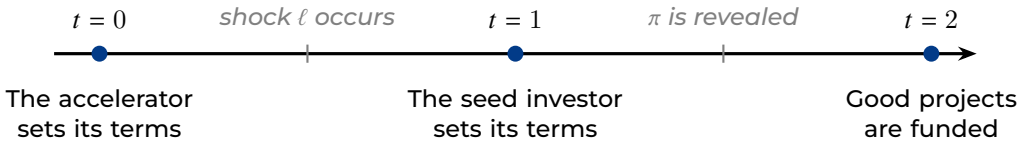
The seed investor

- Invests σ_s after ℓ, before π.
- Makes TIOLI offers.
- Runs an equity round, or a priced or unpriced SAFE.

Experimentation

Covering the liquidity shock ℓ finances an experiment, which reveals the project's value π. The accelerator and the seed investor are the investors able to finance it.

Timing



- The accelerator's contribution is spent before the shock: if it rejects the seed round at $t = 1$, the project is abandoned and it gets nothing.
- If the project is not funded at $t = 2$, it returns nothing, and all the money invested is lost.
- The accelerator's protection comes first. As in YC's post-money SAFE, it is not diluted by the SAFEs signed after it.

We solve the game by backward induction.

Benchmark: no clause, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds no protection. Seed holds a priced SAFE.

Suppose $\ell > \sigma_a$ and $x \in [0, 1]$. The seed investor solves

$$\begin{aligned} \max_{\sigma_s, \rho_s} \quad & \Pi_s = \rho_s\, W(x) - \sigma_s \\ \text{under} \quad & \sigma_s \geq \ell - \sigma_a \qquad (L) \end{aligned}$$

with $\rho_s \in [0, 1]$.

Benchmark: no clause, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds no protection. Seed holds a priced SAFE.

- The seed investor exactly covers the remaining liquidity cost, $\sigma_s^* = \ell - \sigma_a$, and $x^* = I$.
- It takes all of the equity, $\rho_s^* = 1$.
- It earns $\Pi_s^* = W(I) - (\ell - \sigma_a)$, and finances $\ell \leq \sigma_a + W(I)$.
- The accelerator keeps nothing.

Benchmark: no clause, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds no protection. Seed holds a priced SAFE.

The accelerator's contribution is spent before the shock. It solves

$$\max_{\sigma_a \geq 0} \quad \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell - \sigma_a.$$

Benchmark: no clause, priced SAFE

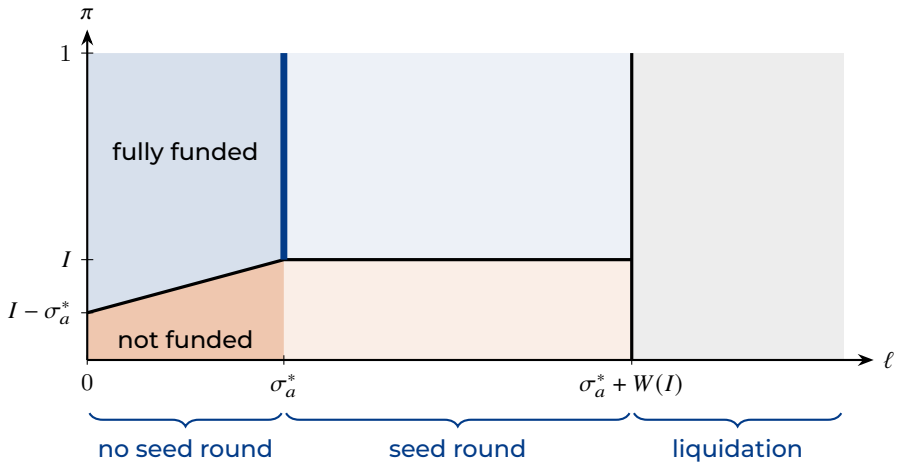
The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds no protection. Seed holds a priced SAFE.

Every optimal contribution σ_a^* is zero or solves

$$W(I)\, g(\sigma_a^*) = \big(1 - G(\sigma_a^*)\big) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell.$$

Benchmark: no clause, priced SAFE



No clause, uncapped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds no protection. Seed holds an uncapped SAFE without discount.

Suppose $\ell > \sigma_a$ and $x \in [0, 1]$. The seed investor solves

$$\begin{aligned} \max_{\sigma_s} \quad & \Pi_s = \int_x^1 \rho_s(\pi)\, (\pi - x)\, f(\pi)\, d\pi - \sigma_s \\ \text{under} \quad & \sigma_s \geq \ell - \sigma_a \end{aligned} \tag{L}$$

with $\rho_s(\pi) = \min\{\sigma_s/(\pi - x),\ 1\}$: the SAFE pays $\min(\sigma_s, \pi - x)$ in state π.

No clause, uncapped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds no protection. Seed holds an uncapped SAFE without discount.

- Since $\min(\sigma_s, \pi - x) = (\pi - x) - (\pi - x - \sigma_s)^+$, the SAFE pays $W(x) - W(x + \sigma_s)$ in expectation, and at most its amount in every state.
- The seed investor loses money at every amount, $\Pi_s = -\int_x^{x+\sigma_s} F(t)\, dt < 0$.
- No shock $\ell > \sigma_a$ obtains a seed round, and the accelerator keeps nothing on these projects.

No clause, uncapped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds no protection. Seed holds an uncapped SAFE without discount.

The accelerator's contribution is spent before the shock. It solves

$$\max_{\sigma_a \geq 0} \quad \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell - \sigma_a.$$

No clause, uncapped SAFE

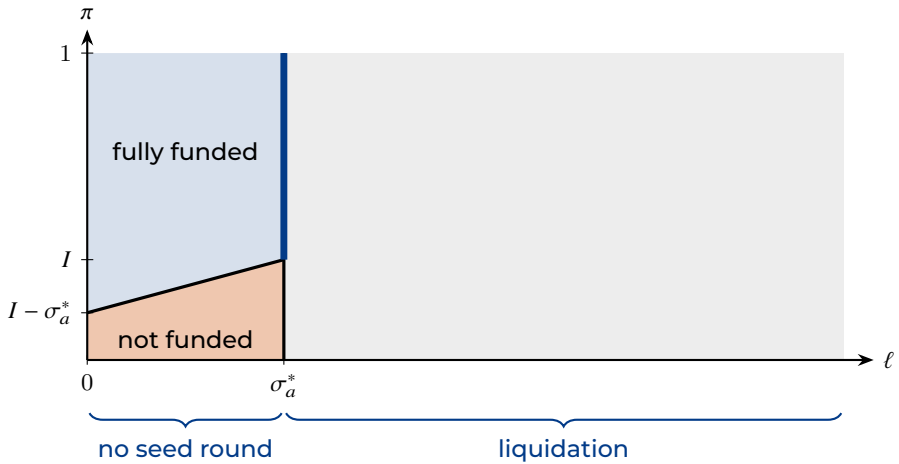
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Accelerator holds no protection. Seed holds an uncapped SAFE without discount.

Every optimal contribution σ_a^* is zero or solves

$$W(I)\, g(\sigma_a^*) = \big(1 - G(\sigma_a^*)\big) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell.$$

It is the benchmark contribution.

No clause, uncapped SAFE



No clause, uncapped SAFE with discount

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds no protection. Seed holds an uncapped SAFE with discount λ.

Suppose $\ell > \sigma_a$ and $x \in [0, 1]$. The seed investor solves

$$\begin{aligned} \max_{\sigma_s, \lambda} \quad & \Pi_s = \int_x^1 \rho_s(\pi)\, (\pi - x)\, f(\pi)\, d\pi - \sigma_s \\ \text{under} \quad & \sigma_s \geq \ell - \sigma_a \end{aligned} \tag{L}$$

with $\rho_s(\pi) = \min\{\sigma_s/(\lambda(\pi - x)),\ 1\}$ and $\lambda \in (0, 1]$: the SAFE pays $\min(\sigma_s/\lambda, \pi - x)$ in state π.

No clause, uncapped SAFE with discount

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds no protection. Seed holds an uncapped SAFE with discount λ.

- Since $\min(\sigma_s/\lambda, \pi - x) = (\pi - x) - (\pi - x - \sigma_s/\lambda)^+$, the SAFE pays $W(x) - W(x + \sigma_s/\lambda)$ in expectation.
- The seed investor exactly covers the remaining liquidity cost, $\sigma_s^* = \ell - \sigma_a$, and $x^* = I$.
- The discount $\lambda^* = \min\{1, (\ell - \sigma_a)/(1 - I)\}$ gives it all of the equity.
- It earns $\Pi_s^* = W(I) - (\ell - \sigma_a)$, and finances $\ell \leq \sigma_a + W(I)$.

No clause, uncapped SAFE with discount

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds no protection. Seed holds an uncapped SAFE with discount λ.

The accelerator's contribution is spent before the shock. It solves

$$\max_{\sigma_a \geq 0} \quad \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell - \sigma_a.$$

No clause, uncapped SAFE with discount

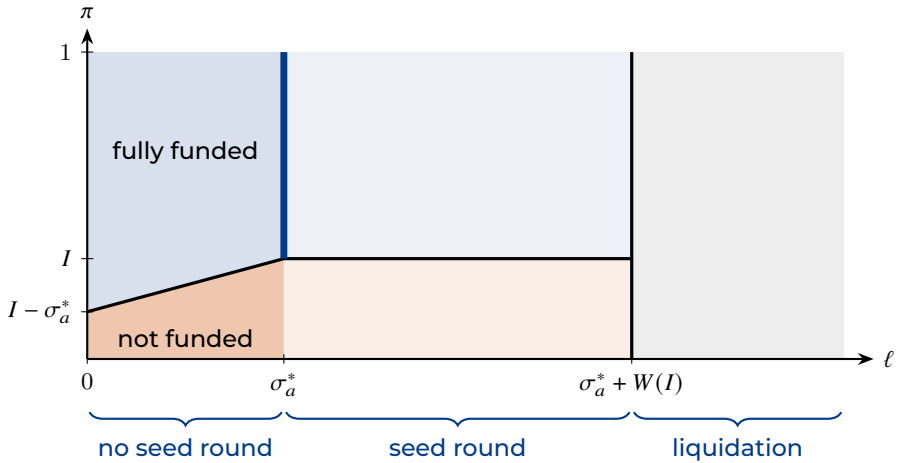
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Accelerator holds no protection. Seed holds an uncapped SAFE with discount λ.

Every optimal contribution σ_a^* is zero or solves

$$W(I)\, g(\sigma_a^*) = \big(1 - G(\sigma_a^*)\big) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell.$$

No clause, uncapped SAFE with discount



No clause, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds no protection. Seed holds a capped SAFE with discount.

Suppose $\ell > \sigma_a$ and $x \in [0, 1]$. The seed investor solves

$$\max_{\sigma_s, V_s, \lambda} \quad \Pi_s = \int_x^1 \rho_s(\pi)\, (\pi - x)\, f(\pi)\, d\pi - \sigma_s$$
$$\text{under} \quad \sigma_s \geq \ell - \sigma_a \qquad (L)$$

with $\rho_s(\pi) = \min\{\sigma_s / \min(V_s, \lambda(\pi - x)),\ 1\}$ and $\lambda \in (0, 1]$.

No clause, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds no protection. Seed holds a capped SAFE with discount.

- The seed investor exactly covers the remaining liquidity cost, $\sigma_s^* = \ell - \sigma_a$, and $x^* = I$.
- The cap $V_s^* = \ell - \sigma_a$ gives it all of the equity.
- It earns $\Pi_s^* = W(I) - (\ell - \sigma_a)$, and finances $\ell \leq \sigma_a + W(I)$.

No clause, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds no protection. Seed holds a capped SAFE with discount.

The accelerator's contribution is spent before the shock. It solves

$$\max_{\sigma_a \geq 0} \quad \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell - \sigma_a.$$

No clause, capped SAFE

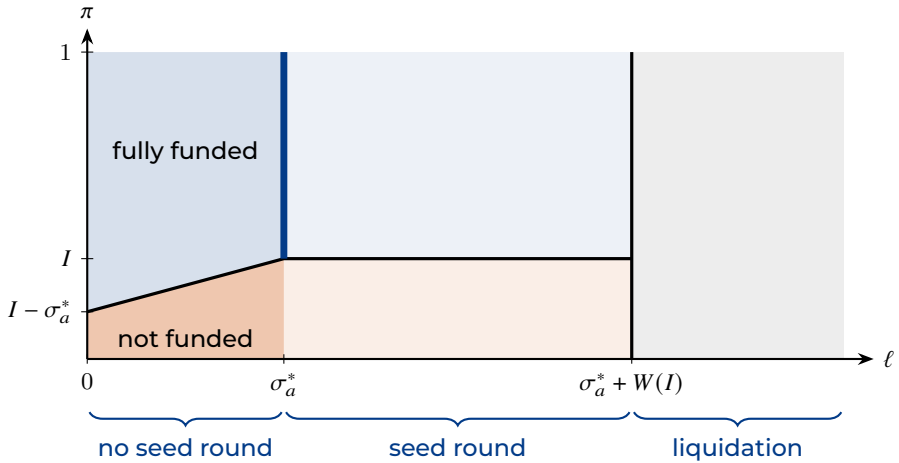
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Accelerator holds no protection. Seed holds a capped SAFE with discount.

Every optimal contribution σ_a^* is zero or solves

$$W(I)\, g(\sigma_a^*) = \big(1 - G(\sigma_a^*)\big) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell.$$

No clause, capped SAFE



Fixed share, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds a fixed share ρ_a. Seed holds a priced SAFE.

Suppose $\ell > \sigma_a$ and $x \in [0, 1]$. The seed investor solves

$$\begin{aligned} \max_{\sigma_s, \rho_s} \quad & \Pi_s = \rho_s\, W(x) - \sigma_s \\ \text{under} \quad & \sigma_s \geq \ell - \sigma_a && (L) \\ & \rho_s \leq 1 - \rho_a && (P) \end{aligned}$$

with $\rho_s \in [0, 1]$.

Fixed share, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds a fixed share ρ_a. Seed holds a priced SAFE.

- The seed investor exactly covers the remaining liquidity cost, $\sigma_s^* = \ell - \sigma_a$, and $x^* = I$.
- The protection binds, $\rho_s^* = 1 - \rho_a$.
- The seed investor earns $\Pi_s^* = (1 - \rho_a)W(I) - (\ell - \sigma_a)$, and finances $\ell \leq \sigma_a + (1 - \rho_a)W(I)$.
- The accelerator keeps $\rho_a W(I)$.

Fixed share, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds a fixed share ρ_a. Seed holds a priced SAFE.

The accelerator solves

$$\max_{\sigma_a, \rho_a} \quad \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell + \int_{\sigma_a}^{\sigma_a + (1-\rho_a)W(I)} \rho_a W(I)\, g(\ell)\, d\ell - \sigma_a.$$

Fixed share, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds a fixed share ρ_a. Seed holds a priced SAFE.

At each σ_a, the optimal share $\rho_a^* \in (0,1)$ is unique and solves

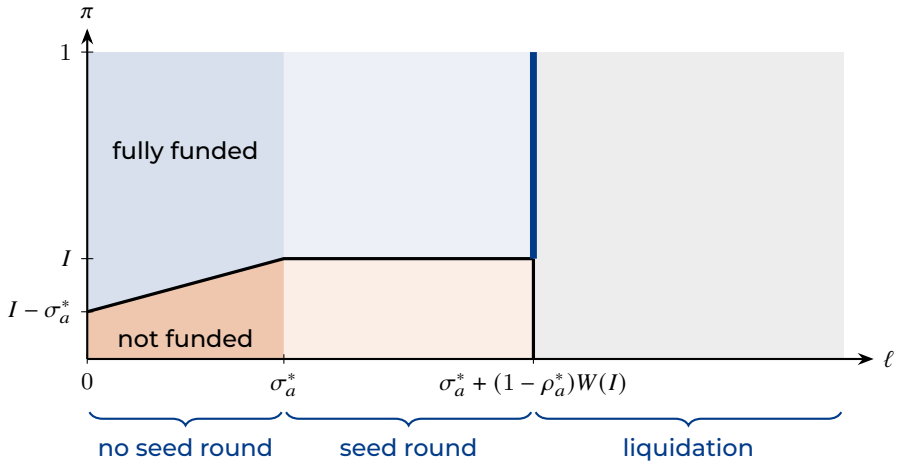
$$G\big(\sigma_a + (1-\rho_a^*)W(I)\big) - G(\sigma_a) = \rho_a^* W(I)\, g\big(\sigma_a + (1-\rho_a^*)W(I)\big).$$

Every optimal contribution σ_a^* is zero or solves

$$(1-\rho_a^*)\, W(I)\, g(\sigma_a^*) + \rho_a^* W(I)\, g\big(\sigma_a^* + (1-\rho_a^*)W(I)\big)$$
$$= \big(1 - G(\sigma_a^*)\big) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell.$$

Compared with the benchmark, the contribution rises if g is higher at the last seed round than at σ_a, and falls otherwise.

Fixed share, priced SAFE



Fixed share, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds a fixed share ρ_a. Seed holds a capped SAFE with discount.

Suppose $\ell > \sigma_a$ and $x \in [0, 1]$. The seed investor solves

$$\begin{aligned} \max_{\sigma_s, V_s, \lambda} \quad & \Pi_s = \int_x^1 \rho_s(\pi)\, (\pi - x)\, f(\pi)\, d\pi - \sigma_s \\ \text{under} \quad & \sigma_s \geq \ell - \sigma_a \end{aligned} \tag{L}$$

with $\rho_s(\pi) = \min\{\sigma_s / \min(V_s, \lambda(\pi - x)),\ \underbrace{1 - \rho_a}_{(P)}\}$ and $\lambda \in (0, 1]$.

Fixed share, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds a fixed share ρ_a. Seed holds a capped SAFE with discount.

- The seed investor obtains exactly what it obtains with a priced SAFE, on the same projects.
- The cap $V_s^* = (\ell - \sigma_a)/(1 - \rho_a)$ gives it the share $1 - \rho_a$ in every state above I.
- The discount plays no role.

Fixed share, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds a fixed share ρ_a. Seed holds a capped SAFE with discount.

The accelerator solves

$$\max_{\sigma_a, \rho_a} \quad \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell + \int_{\sigma_a}^{\sigma_a + (1-\rho_a)W(I)} \rho_a W(I)\, g(\ell)\, d\ell - \sigma_a.$$

Fixed share, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds a fixed share ρ_a. Seed holds a capped SAFE with discount.

At each σ_a, the optimal share $\rho_a^* \in (0,1)$ is unique and solves

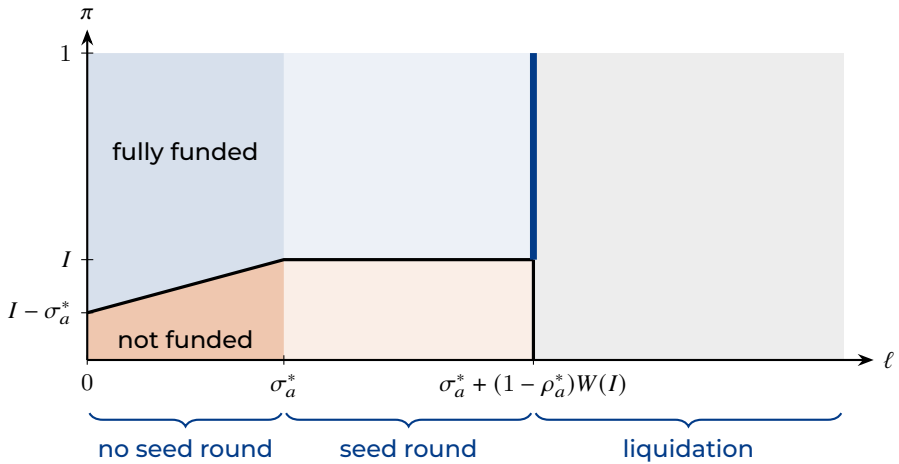
$$G\big(\sigma_a + (1-\rho_a^*)W(I)\big) - G(\sigma_a) = \rho_a^* W(I)\, g\big(\sigma_a + (1-\rho_a^*)W(I)\big).$$

Every optimal contribution σ_a^* is zero or solves

$$(1-\rho_a^*)\, W(I)\, g(\sigma_a^*) + \rho_a^* W(I)\, g\big(\sigma_a^* + (1-\rho_a^*)W(I)\big)$$
$$= \big(1 - G(\sigma_a^*)\big) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell.$$

Compared with the benchmark, the contribution rises if g is higher at the last seed round than at σ_a, and falls otherwise.

Fixed share, capped SAFE



Accelerator's capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds a capped SAFE on its contribution σ_a, with cap V_a and discount λ_a. Seed holds a priced or capped SAFE.

- YC's $125k for 7% is itself a post-money SAFE.
- The accelerator's SAFE claims $\sigma_a / \min(V_a, \lambda_a(\pi - x))$. Times $\pi - x$, this is the larger of the share σ_a / V_a, when the cap binds, and of the value σ_a / λ_a, when the discount binds.
- The protection comes first, and leaves in expectation

$$W\left(x + \frac{\sigma_a}{\lambda_a}\right) - \frac{\sigma_a}{V_a}\, W\left(x + \frac{V_a}{\lambda_a}\right),$$

since $\int_x^1 (\pi - x - y)^+ f(\pi)\, d\pi = W(x + y)$.

Accelerator's capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds a capped SAFE on its contribution σ_a, with cap V_a and discount λ_a. Seed holds a priced or capped SAFE.

- The seed investor takes all the equity the protection leaves, with a priced SAFE ($\rho_s = 1$) or a capped SAFE ($V_s = \ell - \sigma_a$), and invests only its need.
- It finances the shock if and only if

$$\ell - \sigma_a \le W\Big(I + \frac{\sigma_a}{\lambda_a}\Big) - \frac{\sigma_a}{V_a}\, W\Big(I + \frac{V_a}{\lambda_a}\Big).$$

- The accelerator keeps $W(I)$ minus this value on each of these projects.

Accelerator's capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds a capped SAFE on its contribution σ_a, with cap V_a and discount λ_a. Seed holds a priced or capped SAFE.

The accelerator's SAFE never does better than the fixed share, and does as well in all our simulations.

- Its profit depends only on what it keeps at the threshold I, as with a fixed share, since the seed investor invests only its need.
- The SAFE keeps at least $W(I) - W(I + \sigma_a)$: it returns its amount when the project allows it. The fixed share has no such minimum.
- When the accelerator invests, the optimal fixed share keeps 0.125 to 0.168, above this minimum, at most 0.028: capping well is then pricing well.

MFN clause, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). *Seed holds a priced SAFE.*

- Each SAFE claims a share of the equity held before the equity round.
- Each claim is multiplied by $\min\{1,\ (1-\rho_a^0)/(\text{sum of the claims})\}$: kept if the claims fit in $1-\rho_a^0$, scaled down to fill it otherwise.
- The seed investor claims ρ_s. The amount m converts at $\min(\sigma_s/\rho_s,\ \pi - x)$ and claims $m/\min(\sigma_s/\rho_s,\ \pi - x)$.
- If $\pi - x \geq \sigma_s/\rho_s$, the claims sum to $\rho_s(\sigma_s + m)/\sigma_s$, and the seed investor keeps

$$\min\left\{\rho_s,\ \rho_s\,\frac{1-\rho_a^0}{\rho_s(\sigma_s+m)/\sigma_s}\right\} = \min\left\{\rho_s,\ \frac{(1-\rho_a^0)\,\sigma_s}{\sigma_s+m}\right\}.$$

- If $\pi - x < \sigma_s/\rho_s$, they sum to $\rho_s + m/(\pi - x)$, and it keeps

$$\min\left\{\rho_s,\ \rho_s\,\frac{1-\rho_a^0}{\rho_s+m/(\pi-x)}\right\} = \min\left\{\rho_s,\ \frac{(1-\rho_a^0)\,\rho_s(\pi-x)}{\rho_s(\pi-x)+m}\right\}.$$

MFN clause, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a priced SAFE.

Suppose $\ell > \sigma_a$ and $x \in [0, 1]$. The seed investor solves

$$\begin{aligned} \max_{\sigma_s, \rho_s} \quad & \Pi_s = \int_x^1 \rho_s(\pi)\,(\pi - x)\, f(\pi)\, d\pi - \sigma_s \\ \text{under} \quad & \sigma_s \geq \ell - \sigma_a \qquad (L) \end{aligned}$$

with

$$\rho_s(\pi) = \begin{cases} \min\{\rho_s,\ (1-\rho_a^0)\,\sigma_s/(\sigma_s + m)\} & \text{if } \pi - x \geq \sigma_s/\rho_s, \\ \min\{\rho_s,\ (1-\rho_a^0)\,\rho_s(\pi - x)/(\rho_s(\pi - x) + m)\} & \text{if } \pi - x < \sigma_s/\rho_s, \end{cases}$$

as derived on the previous slide, where the second term of each $\min$ is the protection (MFN).

MFN clause, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a priced SAFE.

- The amount m converts at $\min(\sigma_s/\rho_s,\ \pi - x)$.
- The seed investor's share never exceeds $(1 - \rho_a^0)\sigma_s/(\sigma_s + m)$, and, if $m > 0$, is strictly below it when $\pi - x < \sigma_s/\rho_s$.
- The seed investor finances no shock with $\ell - \sigma_a + m > (1 - \rho_a^0)W(I)$.
- If $m > 0$, it earns strictly less than a capped SAFE gives it.

MFN clause, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). *Seed holds a priced SAFE.*

With $\varphi(\ell) = \Pi_s^*(\ell) + \int_{x^*(\ell)}^{I} F(x)\, dx$, the accelerator solves

$$\max_{\sigma_a, \rho_a^0, m \le \sigma_a} \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell + \int_{\sigma_a}^{\bar{\ell}} \big[W(I) - (\ell - \sigma_a) - \varphi(\ell)\big] g(\ell)\, d\ell - \sigma_a,$$

with $\bar{\ell}$ the largest shock the seed investor finances.

MFN clause, priced SAFE

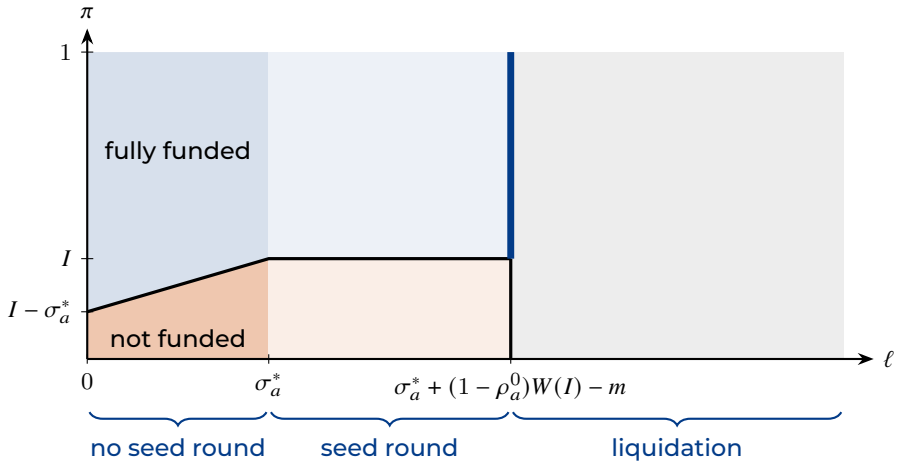
The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a priced SAFE.

If the optimal fixed share comes with a contribution, every optimum has $m^* > 0$. If the seed investor finances all the shocks between σ_a^* and $\bar{\ell}$, at given (ρ_a^0, m) with $m < \sigma_a^*$, the contribution solves

$$\left[W(I) - (\bar{\ell} - \sigma_a^*)\right] g(\bar{\ell}) = \left(1 - G(\bar{\ell})\right) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell + \int_{\sigma_a^*}^{\bar{\ell}} \varphi(\ell)\, g'(\ell)\, d\ell.$$

MFN clause, priced SAFE



MFN clause, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a capped SAFE with discount.

- Each SAFE claims a share of the equity held before the equity round.
- Each claim is multiplied by $\min\{1,\ (1-\rho_a^0)/(\text{sum of the claims})\}$: kept if the claims fit in $1-\rho_a^0$, scaled down to fill it otherwise.
- Both SAFEs convert at $\min(V_s, \lambda(\pi - x))$: the seed investor claims $\sigma_s/\min(V_s, \lambda(\pi - x))$, and the claims sum to $(\sigma_s + m)/\min(V_s, \lambda(\pi - x))$.
- The seed investor keeps

$$\min\left\{\frac{\sigma_s}{\min(V_s, \lambda(\pi - x))},\ \frac{\sigma_s}{\min(V_s, \lambda(\pi - x))} \cdot \frac{(1-\rho_a^0)\min(V_s, \lambda(\pi - x))}{\sigma_s + m}\right\},$$

  that is $\min\{\sigma_s/\min(V_s, \lambda(\pi - x)),\ (1-\rho_a^0)\,\sigma_s/(\sigma_s + m)\}$.

MFN clause, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a capped SAFE with discount.

Suppose $\ell > \sigma_a$ and $x \in [0,1]$. The seed investor solves

$$\max_{\sigma_s, V_s, \lambda} \quad \Pi_s = \int_x^1 \rho_s(\pi)\,(\pi - x)\, f(\pi)\, d\pi - \sigma_s$$
$$\text{under} \quad \sigma_s \geq \ell - \sigma_a \qquad (L)$$

with $\rho_s(\pi) = \min\{\sigma_s / \min(V_s, \lambda(\pi - x)),\ \underbrace{(1-\rho_a^0)\,\sigma_s/(\sigma_s + m)}_{(MFN)}\}$ and $\lambda \in (0,1]$, as derived on the previous slide.

MFN clause, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a capped SAFE with discount.

- The seed investor finances the shock if and only if $\ell - \sigma_a + m \le (1 - \rho_a^0) W(I)$.
- The clause binds, $\rho_s^* = (1 - \rho_a^0)\sigma_s^* / (\sigma_s^* + m)$, with the cap $V_s^* = (\sigma_s^* + m)/(1 - \rho_a^0)$, and the discount plays no role.
- The seed investor invests beyond its need, $\sigma_s^* > \ell - \sigma_a$ and $x^* < I$, if

$$(1 - \rho_a^0)\Big[\frac{m\, W(I)}{(\ell - \sigma_a + m)^2} + \frac{(\ell - \sigma_a)\big(1 - F(I)\big)}{\ell - \sigma_a + m}\Big] > 1,$$

  which holds on the smallest rounds.

MFN clause, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a capped SAFE with discount.

With $\varphi(\ell) = \Pi_s^*(\ell) + \int_{x^*(\ell)}^{I} F(x)\, dx$, the accelerator solves

$$\max_{\sigma_a, \rho_a^0, m \le \sigma_a} \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell + \int_{\sigma_a}^{\bar{\ell}} \big[W(I) - (\ell - \sigma_a) - \varphi(\ell)\big] g(\ell)\, d\ell - \sigma_a,$$

with $\bar{\ell}$ the largest shock the seed investor finances, $\bar{\ell} = \sigma_a + (1 - \rho_a^0) W(I) - m$.

MFN clause, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

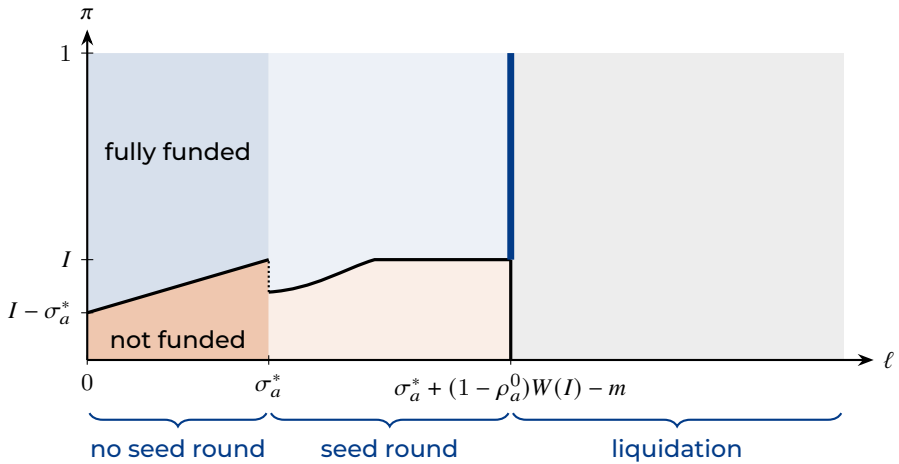
Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a capped SAFE with discount.

If the optimal fixed share comes with a contribution, every optimum has $m^* > 0$. At given (ρ_a^0, m) with $m < \sigma_a^*$, the contribution solves

$$\left(\rho_a^0 W(I) + m\right) g(\bar{\ell}) = \left(1 - G(\bar{\ell})\right) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell + \int_{\sigma_a^*}^{\bar{\ell}} \varphi(\ell)\, g'(\ell)\, d\ell,$$

with $\bar{\ell} = \sigma_a^* + (1 - \rho_a^0) W(I) - m$ the last shock the seed investor finances, on which the accelerator keeps $\rho_a^0 W(I) + m$.

MFN clause, capped SAFE



MFN clause with the accelerator's SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause whose fixed part is its own SAFE, on $\sigma_a - m$. Seed holds a capped SAFE.

The fixed part of the clause can be a share or the accelerator's own SAFE: numerically, it makes no difference.

- The seed investor and the MFN component share what the accelerator's SAFE leaves, and the seed investor still prefers the capped SAFE.
- The fixed share is the limit of a SAFE of cap $(\sigma_a - m)/\rho_a^0$ as $m \to \sigma_a$.
- The value $(\sigma_a - m)/\lambda_a$ kept in the low states changes the seed investor's answer only on the small rounds where it invests beyond its need.
- In all configurations, it raises the accelerator's profit by at most $2 \cdot 10^{-5}$, the numerical precision.

Summary

Accelerator	Seed investor	Seed round	Overinvestment	Accelerator keeps if seed
No clause	Priced SAFE	yes	no	nothing
No clause	Uncapped SAFE	no	–	–
No clause	Uncapped SAFE with discount	yes	no	nothing
No clause	Capped SAFE with discount	yes	no	nothing
Fixed share	Priced SAFE	yes	no	$\rho_a W(I)$
Fixed share	Capped SAFE with discount	yes	no	$\rho_a W(I)$
Own SAFE	Any SAFE	yes	no	$\geq W(I) - W(I + \sigma_a)$
MFN clause	Priced SAFE	yes	possible	$\geq \rho_a^0 W(I)$
MFN clause	Capped SAFE with discount	yes	small rounds	$\geq \rho_a^0 W(I)$

Accelerator's profit: MFN clause $>$ fixed share $\geq$ own SAFE, and fixed share $>$ no clause, the first inequality when the optimal fixed share comes with a contribution.

The MFN clause beats any fixed share

Accelerator	Accelerator's profit
Fixed share	$B(\sigma_a^P) + \rho_a^P W(I)\left[G\left(\sigma_a^P + (1-\rho_a^P)W(I)\right) - G(\sigma_a^P)\right]$
MFN clause	$B(\sigma_a^M) + \int_{\sigma_a^M}^{\sigma_a^M + (1-\rho_a^{0M})W(I) - m^M} \left[W(I) - (\ell - \sigma_a^M) - \varphi(\ell)\right] g(\ell)\, d\ell$

P and M denote the optimal fixed share and the optimal MFN clause,
$B(\sigma_a) = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell - \sigma_a$ the profit on the projects financed alone, and
$\varphi(\ell) = \Pi_s^* + \int_{x^*}^{I} F$ the seed investor's profit plus the money lost beyond its need.

- When the optimal fixed share comes with a contribution, the MFN clause gives strictly more, with either SAFE.
- The MFN amount cannot exceed the contribution: without a contribution, the clause is a fixed share.
- Which SAFE gives the accelerator more under the MFN clause is open.

A small MFN component pays most on small rounds

Start from the optimal fixed share, with $\sigma_a^P > 0$, and add a small m.

- The seed investor's share never exceeds $(1 - \rho_a^0)\sigma_s/(\sigma_s + m)$, whatever its SAFE.
- On round $\ell - \sigma_a$, the value the accelerator keeps rises by at least

$$\frac{(1 - \rho_a^0)\, m\, W(I)}{\ell - \sigma_a + m}.$$

- The seed investor stops financing only rounds within $O(m)$ of the ends of its zone.

Divided by m, the gain is $(1 - \rho_a^0)W(I) \int g(\ell)/(\ell - \sigma_a)\, d\ell$, unbounded near $\ell = \sigma_a$, and the loss stays bounded: $m^* > 0$.

A project is fully funded above its threshold

The probability of full funding is

$$\int_0^{\sigma_a} \big(1 - F(I + \ell - \sigma_a)\big)\, g(\ell)\, d\ell + \int_{\sigma_a}^{\sigma_a + (1-\rho_a^0)W(I) - m} \big(1 - F(x^*)\big)\, g(\ell)\, d\ell.$$

- The seed investor finances $\sigma_a < \ell \leq \sigma_a + (1 - \rho_a^0)W(I) - m$, where the accelerator keeps $\rho_a^0 W(I) + m$ on the last project. A fixed share is $m = 0$, no clause is $\rho_a^0 = m = 0$.
- On a project the accelerator finances alone, the unused cash lowers the threshold below I.
- The seed investor covers only its need, $x^* = I$, except on small rounds under the MFN clause, $x^* < I$.
- A dollar more of contribution lowers the thresholds and moves the seed investor's zone up: full funding rises.

The fixed share moves full funding either way

Moving from no clause to the optimal fixed share changes the probability of full funding by

$$\underbrace{\int_{\sigma_a^B}^{\sigma_a^P} \left[\int_0^s f(I+\ell-s)\, g(\ell)\, d\ell + \big(1-F(I)\big) g\big(s+W(I)\big) \right] ds}_{\text{contribution}} - \underbrace{\big(1-F(I)\big)\Big[G\big(\sigma_a^P + W(I)\big) - G\big(\sigma_a^P + (1-\rho_a^P)W(I)\big)\Big]}_{\text{last project} > 0}.$$

- The contribution term has the sign of $\sigma_a^P - \sigma_a^B$.
- If the fixed share does not raise the contribution, full funding falls.

The MFN clause moves full funding either way

With a capped SAFE, moving from the fixed share to the MFN clause changes the probability of full funding by

$$\underbrace{\int_{\sigma_a^P}^{\sigma_a^M} \Big[\int_0^s f(I+\ell-s)\, g(\ell)\, d\ell + \big(1-F(I)\big) g\big(s+(1-\rho_a^P)W(I)\big) \Big] ds}_{\text{contribution}}$$

$$+ \underbrace{\big(1-F(I)\big)\Big[G(\bar{\ell}^M) - G\big(\sigma_a^M + (1-\rho_a^P)W(I)\big)\Big]}_{\text{last project}} + \underbrace{\int_{\sigma_a^M}^{\bar{\ell}^M} \big(F(I)-F(x^*)\big) g(\ell)\, d\ell}_{\text{beyond the need } \geq 0},$$

- $\bar{\ell}^M = \sigma_a^M + (1-\rho_a^{0M})W(I) - m^M$ is the last shock the seed investor finances under the MFN clause.
- The contribution term has the sign of $\sigma_a^M - \sigma_a^P$, the last-project term the sign of $\rho_a^P W(I) - \rho_a^{0M} W(I) - m^M$.
- Only the investment beyond the need is signed.

A seed round covers at most $W(I) < I$

Accelerator	Probability of a seed round
No clause	$G\big(\sigma_a^B + W(I)\big) - G(\sigma_a^B)$
Fixed share	$G\big(\sigma_a^P + (1-\rho_a^P)W(I)\big) - G(\sigma_a^P)$
MFN clause, capped SAFE	$G\big(\sigma_a^M + (1-\rho_a^{0M})W(I) - m^M\big) - G(\sigma_a^M)$
MFN clause, priced SAFE	below the capped SAFE

- Whatever the protection, the seed investor finances a shock only if $\ell - \sigma_a \leq W(I) < I$.
- A protection keeps $\rho_a^0 W(I) + m$ for the accelerator on the last project, and shortens the zone by as much.
- A different contribution slides the zone: the probability rises or falls with the slope of g.

The MFN clause moves seed rounds either way

With a capped SAFE, moving from the fixed share to the MFN clause changes the probability of a seed round by

$$\underbrace{\left[G\left(\sigma_a^M + (1-\rho_a^P)W(I)\right) - G(\sigma_a^M)\right] - \left[G\left(\sigma_a^P + (1-\rho_a^P)W(I)\right) - G(\sigma_a^P)\right]}_{\text{contribution}}$$

$$+ \underbrace{G(\bar{\ell}^M) - G\left(\sigma_a^M + (1-\rho_a^P)W(I)\right)}_{\text{last project}}.$$

- The last-project term has the sign of $\rho_a^P W(I) - \rho_a^{0M} W(I) - m^M$.
- The contribution term has no sign.

Total surplus: four losses from the first best

The total surplus is the sum of all payoffs. A covered project with threshold x gives $W(I) - \ell - \int_x^I F$, an abandoned one $-\sigma_a$. The seed investor finances up to $\bar{\ell} = \sigma_a + (1 - \rho_a^0)W(I) - m$.

$$TS = \underbrace{\int_0^{W(I)} \big(W(I) - \ell\big) g(\ell)\, d\ell}_{\text{first best}} - \underbrace{\int_0^{\sigma_a} \int_{I+\ell-\sigma_a}^{I} F\; g(\ell)\, d\ell}_{L_{acc}} - \underbrace{\int_{\bar{\ell}}^{W(I)} \big(W(I) - \ell\big) g(\ell)\, d\ell}_{L_{cov}}$$

$$- \underbrace{\int_{\sigma_a}^{\bar{\ell}} \int_{x^*}^{I} F\; g(\ell)\, d\ell}_{L_{over}} - \underbrace{\sigma_a \big(1 - G(\bar{\ell})\big)}_{L_{sunk}}$$

- L_{acc} and L_{over}: money beyond the need is lost on abandoned projects and funds projects worth less than I.
- L_{cov}: shocks left uncovered below $W(I)$, or covered above it.
- L_{sunk}: the contribution spent on abandoned projects. The first best has $\sigma_a = 0$.

The fixed share moves total surplus either way

Moving from no clause to the optimal fixed share changes total surplus by

$$\underbrace{\int_{\sigma_a^P}^{\sigma_a^B} \Big[\big(1 - G(s + W(I))\big) + \int_0^s F(I + \ell - s)\, g(\ell)\, d\ell\Big] ds}_{\text{contribution}} - \underbrace{\int_{\sigma_a^P + (1-\rho_a^P)W(I)}^{\sigma_a^P + W(I)} \big(\sigma_a^P + W(I) - \ell\big) g(\ell)\, d\ell}_{\text{last project} > 0}.$$

- Without clause, total surplus falls with the contribution.
- The contribution term has the sign of $\sigma_a^B - \sigma_a^P$.
- The fixed share stops the seed investor $\rho_a^P W(I)$ before $\sigma_a^P + W(I)$.

The MFN clause moves total surplus either way

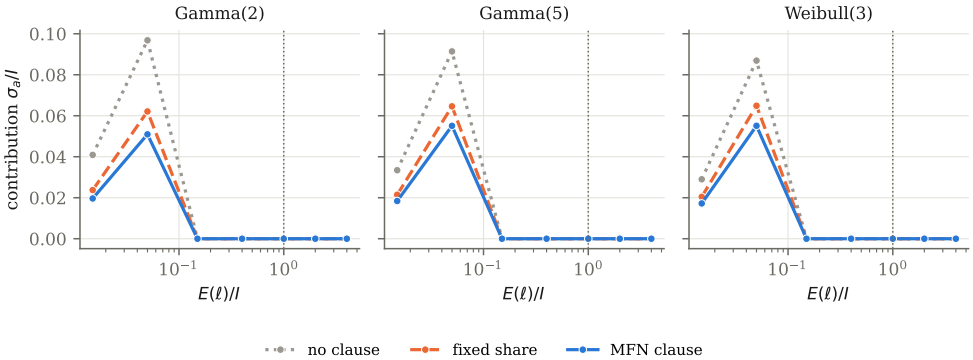
With a capped SAFE, moving from the fixed share to the MFN clause changes total surplus by

$$\underbrace{\int_{\sigma_a^P}^{\sigma_a^M} \Big[\rho_a^P W(I)\, g\big(s+(1-\rho_a^P)W(I)\big) - \big(1-G(s+(1-\rho_a^P)W(I))\big) - \int_0^s F\, g\, d\ell\Big] ds}_{\text{contribution}}$$

$$+ \underbrace{\int_{\sigma_a^M+(1-\rho_a^P)W(I)}^{\bar{\ell}^M} \big(\sigma_a^M + W(I) - \ell\big) g(\ell)\, d\ell}_{\text{last project}} - \underbrace{\int_{\sigma_a^M}^{\bar{\ell}^M} \int_{x^*}^{I} F\, g(\ell)\, d\ell}_{\text{beyond the need} \geq 0}.$$

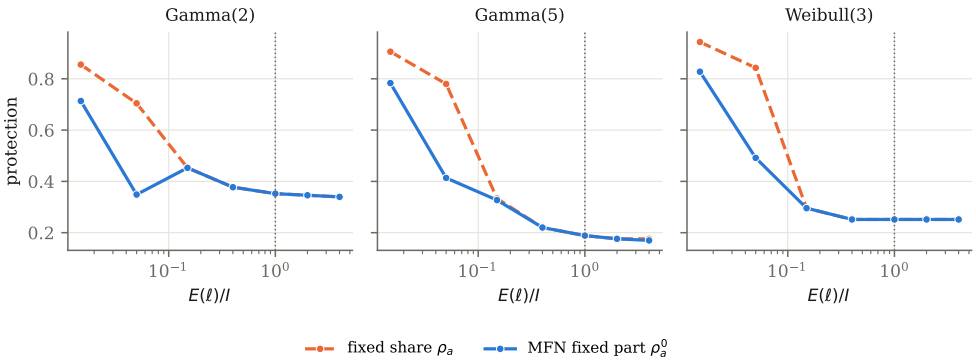
- The last-project term has the sign of $\rho_a^P W(I) - \rho_a^{0M} W(I) - m^M$, the contribution term no sign.
- The investment beyond the need raises full funding, but lowers total surplus.

The accelerator invests only when liquidity needs are small



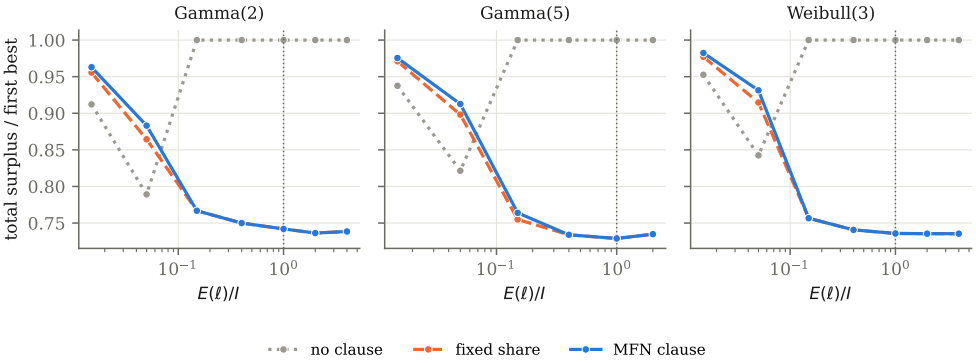
Software, $E(\ell)/I \leq 0.05$: the accelerator invests, less with a protection, and almost all of its money goes through the clause. From $E(\ell)/I = 0.15$ on, it invests nothing under any protection.

Without a contribution, the protection is a fixed share



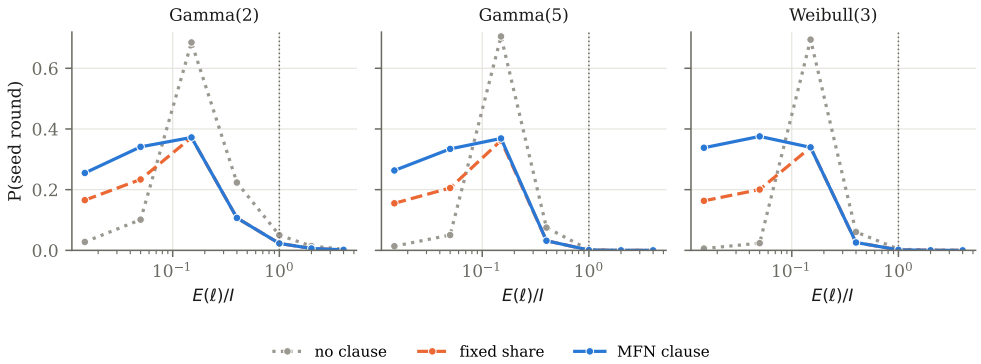
With $\sigma_a = 0$, the MFN amount is zero, $m \leq \sigma_a$, and the accelerator keeps a fixed share of a project it does not fund.

Welfare: total surplus



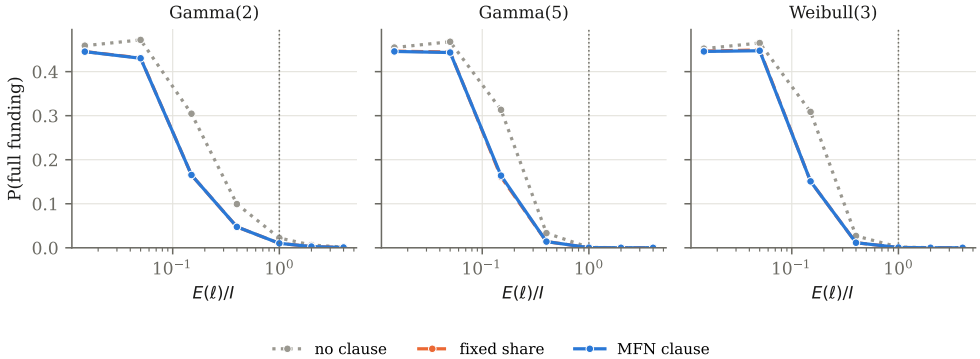
Software: MFN clause > fixed share > no clause. Beyond: no clause reaches the first best, and the protection loses about a quarter of it.

Which projects survive: seed rounds



Software: the protection lowers the contribution and leaves more shocks to the seed investor. Beyond: it only shortens the seed investor's zone.

Which projects survive: full funding



Robustness: another distribution of π

With a Beta$(2, 2)$ value on $[0, 1.8]$ instead of a uniform, nothing changes.

- Software: MFN clause $>$ fixed share $>$ no clause, all the money through the clause, total surplus 0.82–0.97 of the first best with the clause.
- Deep tech: the accelerator invests nothing, keeps a fixed share of 0.18 to 0.42, and the total surplus falls to 0.74–0.76 of the first best.

Robustness: the founders keep a stake

The founders keep ρ_e before the seed round, and the seed investor may take at most $1 - \rho_e - \rho_a$.

- The seed investor's answers hold with $\rho_e + \rho_a$ instead of ρ_a, and its cap $\sigma_s/(1-\rho_e)$ now exceeds its amount.
- Without protection, the first best is no longer reached: the seed investor covers only $\ell - \sigma_a \le (1-\rho_e)W(I)$.
- With $\rho_e = 0.5$ and $E(\ell)/I = 0.05$, the optimal fixed share brings no money, while the MFN clause keeps the accelerator investing: total surplus 0.79–0.87 with the clause, 0.62–0.67 with the fixed share.

Takeaways

Main result

When the accelerator invests, an MFN clause beats any fixed share because it discriminates between rounds.

- The seed investor answers by investing beyond its need on the smallest rounds, and the fixed share survives only to curb this answer.
- Against the clause, the seed investor prefers an unpriced SAFE.
- The accelerator's money is spent: it keeps value only on the shocks it covers alone, and invests only when liquidity needs are small.
- Beyond software, it invests nothing, and its fixed share costs about a quarter of the total surplus, while no protection reaches the first best.
- A capped SAFE for the accelerator never does better than a fixed share, and does as well in our simulations.

Thank you

Protection and Dilution:
SAFEs and MFN Clauses in Seed Financing

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